

20260719 - IBD Shop Similarity Analysis

Archit. - Innovation

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1 Confirmed Anomalies

- [Plots](#)¹
- Insights:

1. https://nielsenenterprise-my.sharepoint.com/:f/g/personal/glen_sequeira_nielseni_com/IgDajQydybB1Qa0Ao0WVJ92jAZv4Je5lQRUkNlienlj7Z1k?e=3klrTR

2 Similarity Logic:

1. Compare the target shop only against shops belonging to the same IBD
2. Fill missing weeks after the first observation with **0** to create continuous histories
3. Remove week **2407** from similarity computation
4. Apply a log transformation and calculate week-over-week change to focus on trend shape

$$L_t = \log(1 + X_t)$$

$$D_t = L_t - L_{t-1}$$

5. Compare only the weeks available in both shops and require a minimum overlap

$$Distance = \sqrt{\sum_{t=1}^n (D_t^{target} - D_t^{candidate})^2}$$

6. Measure the difference between the two transformed trend series using Euclidean distance & adjust the distance by the number of overlapping weeks

$$Normalized\ Distance = \frac{Distance}{\sqrt{n}}$$

7. Sort candidate shops by Normalized Distance: Lower score = higher similarity

3 Implementation of *get_peer_shops* tool:

3.1 Metrics:

- Mean
- Median
- Current shop value
- Current vs Peer Mean %age

$$\frac{(\text{Current Value} - \text{Peer Mean})}{\text{Peer Mean}} \times 100$$

- Peer Latest Change percentage

$$\frac{(\text{Current Peer Mean} - \text{Previous Peer Mean})}{\text{Previous Peer Mean}} \times 100$$

- Note

3.2 Issues:

- The scale of sales of all the peer shops will be different
- Taking all shops in an IBD doesn't make sense

4 Proposed Solution:

4.1 Metrics

- Metrics of the peers

target_growth_percentage:

- target shop's percentage movement from 2406 to 2407

$$\text{target_growth_pct} = 100 \times \frac{x_{2407} - x_{2406}}{\max(x_{2406}, 1)}$$

- peer growth percentage median

1. For every selected peer compute $r_i = \frac{x_{2407}^{(i)} - x_{2406}^{(i)}}{\max(x_{2406}^{(i)}, 1)}$.

2. Take the median m .

3. Report as percentage:

$$\text{peer_growth_pct_median} = 100 \times m$$

- Expected value

$$\text{expected_value}_{2407} = x_{2406} \times (1 + m)$$

Threshold or Choice of N:

- Experiments could be performed with N = 3 or 5 or 7